The impact of product management on SME performance

Evidence from Canadian firms

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Abstract

Purpose – The purpose of this paper is to examine the impact of product management as a set of organizational capabilities. It aims to investigate product management as a set of boundary spanning capabilities, by empirically relating these to firm performance.

Design/methodology/approach – A measurement instrument is developed and validated based on the extant product management literature. Using a heterogeneous sample of 63 Atlantic Canadian SMEs in the manufacturing and professional/technical services sectors, data are collected to test the survey instrument and establish preliminary construct validity.

Findings – Both firm performance and product management measures demonstrate internal consistency. Several product management sub-constructs demonstrated reliability and in some cases validity, substantiating the product management literature. These included product pricing, sales support and forecasting.

Research limitations/implications – This research builds upon the literature and indicates that a relationship exists between product management capability and firm performance. This leads to the conclusion that product management, as a set of boundary spanning firm capabilities, warrants future research with a larger more homogeneous population. Limitations include geographic bias, treating the population as homogeneous and lack of relationship to established firm orientations.

Practical/implications – This research may have practical significance and managerial implications, based on the relationship between product management capabilities and firm performance. This could lead to an increased understanding of how to allocate scarce resources in order to improve performance.

Originality/value – The paper introduces the concept of boundary spanning, product management capabilities and their relationship to firm performance, by providing preliminarily validation of a measurement scale for product management capabilities of small to medium-sized enterprises.

Keywords Product management, Small to medium-sized enterprises, Company performance, Firm orientation, Boundary spanning, Marketing, Canada

Paper type Research paper

1. Introduction

Product management as an organizational concept has been around for over a century in various forms (Katsanis and Pitta, 1995). This boundary spanning capability has a long history of management practice stemming back to the late nineteenth century, with the organizational structure eventually formalized by Proctor and Gamble in the

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early 1930s (Katsanis and Pitta, 1995; Sands, 1979; Dominguez, 1971). This system, which treated the product as the focal point of the management structure, became the standard in most large consumer product organizations and many industrial companies in the 1960s (Sands, 1979; Buell, 1975).

The classic product management model involved the integration of all the functions required for the successful creation, production and marketing of a product line. The strength of this management system was its ability to continuously match internal resources with external requirements, in order to optimize the market performance of individual products or product lines. In the late 1960s, largely driven by the popularization of the study of management (Drucker, 1954), firms became more sophisticated in their approach to the enterprise, eventually relegating product management to a number of subordinate, specialized functions within the organization. These came to include such positions as brand manager, category manager (Katsanis and Pitta, 1999) and technical product manager.

Given the broad nature of the product management concept and its lack of fit within the conventional divisions of management research, few academics have devoted much attention to this field in the modern era of business research. Recent literature can be predominantly classified into three distinct areas; practitioner focused (Gorchels, 2005; Connolly, 2002; Berek, 1998), job function centered (Katsanis and Pitta, 1999; Dawes and Patterson, 1988; McDaniel and Gray, 1980; Venkatesh and Wilemon, 1976; Gorchels, 2003) or based on the agent-firm relationship (Cossé and Swan, 1983; Gemmill, 1972; Giese and Wiesenberger, 1982; Murphy and Gorchels, 1996; Cummings *et al.*, 1989; Lysonski *et al.*, 1995; Strieter *et al.*, 1999). Although each area has benefited from various formal empirical studies (Dawes and Patterson, 1988; Cummings *et al.*, 1989; Lysonski, 1985; Lysonski and Woodside, 1989; Tyagi and Sawhney, 2010; Katsanis, 2006), none have examined product management as a set of organizational capabilities or their relationship to firm performance.

There is however general agreement within the literature on the scope of activities that comprise the aggregate product management responsibilities within the firm and which areas require organizational focus in order to optimize performance. Thus, the premise of this research is that these activities impact firm performance. This study adds to the literature by taking an unconventional approach, whereby it examines product management as a set of boundary spanning organizational capabilities. This product (or service) management capability involves the coordination not only across the traditional boundaries of the firm, but at the interface between the firm and the market (Lysonski, 1985). It goes beyond recent research (Tyagi and Sawhney, 2010), which investigates the organizational determinants of product management, but stops short of investigating product management as a firm capability.

This paper first reviews the extant literature on product management, providing content validity. Next, a synthesized measurement instrument consisting of 32 questions is proposed, covering the breadth of the product management capabilities of the firm. Research methods are then discussed and empirical results analyzed, based on a pilot study of Atlantic Canadian SMEs in the manufacturing and technical service sectors.

For the purposes of this paper, the term "product" will be used to describe both products and services.

2. Conceptual background – product management

Product management (not to be confused with product development, project management or product marketing) had its formal beginnings in the 1930s when Procter and Gamble (P&G) first implemented the role within its divisions as a separate business function (Sands, 1979; Lysonski, 1985; Eckles and Novotny, 1984; Wood and Tandon, 1994). This new function was delegated to an individual who was entirely responsible for the development, production and marketing of a particular product line, integrating all of the functions required for its successful creation and marketing. This "conception through to purchase" system was adapted widely until the 1970s when multifunctional teams began to take over responsibility for the entire product life cycle, replacing individual product managers (Katsanis *et al.*, 1996). While in large organizations this improved the firm's market orientation, it had the unintended effect of dispersing product responsibility throughout the organization.

Several issues arise within the literature, which need to be explored in order to fully understand the role of product management within the business enterprise. According to Cummings (1984) the product management literature can be divided into two areas; namely the background and characteristics of product managers themselves (Cossé and Swan, 1983; Gemmill, 1972) and the organizational characteristics of the position. This research will focus mostly on the latter.

Boundary spanning is the subject of a substantial portion of the literature on the role of product managers and is considered a critical element linking information to the organization (Katsanis and Pitta, 1995; McDaniel and Gray, 1980; Lysonski, 1985; Lysonski and Woodside, 1989; Wood and Tandon, 1994; Lysonski *et al.*, 1988). Lysonski (1985, p. 26) defines boundary spanning as "the interface between the firm and its market environment, as well as between departments in the firm". Product managers thus perform two functions, one as a filter (drawing inferences from information and presenting a homogeneous interpretation) and the other as a facilitator (presenting meaningful information for decision making) (Wood and Tandon, 1994). Paradoxically, it seems that product managers spend most of their time and resources on internal boundary spanning activities, even though external activities are deemed to be more critical to the firm's long-term performance (Gorchels, 2005; Tyagi and Sawhney, 2010).

One of the classic boundary spanning activities is the nurturing of shared cross-functional understanding of customer needs. It requires a culture that breaks down the barriers between such functional areas as sales, marketing, engineering and R&D to solve customer problems. This can be particularly important for SMEs, since the ability to deliver superior products and services is the foundation of many small firm niche strategies (Pelham, 1997). When looking at the growth of SMEs relative to organizational capabilities, Chaston and Mangles (1997) established a relationship between firm growth and internal capabilities. Although several of these capabilities were not directly related to product management (such as investing in productivity assets), several were. What emerged was the critical importance of adopting a balanced approach, believing that emphasis on a single dimension of capability presents a significant risk to the firm, leading to a weakening of future growth (Chaston and Mangles, 1997).

Smallbone *et al.* (1995) on the other hand concentrated on strategies and management actions that affect growth. Related to the product management, they identified six variables that reflect the firm's product and market development. These include:

(1) identifying and responding to new market opportunities;
(2) increasing the breadth of customer base;
(3) broadening or developing a different range of products;
(4) product innovation;
(5) improving competitiveness (for example cost reduction, sales efforts, etc.); and
(6) competitive stance (for example cost-leadership versus differentiation).

These most consistently distinguished high growth firms from other firms in their study (Smallbone *et al.*, 1995).

Although the product manager as an entity has lost much of its luster (Wood and Tandon, 1994), companies must still rely on a system to coordinate the many activities in developing and marketing their products or services. Product management thus should be viewed as a holistic, interdependent, boundary-spanning system (Katsanis and Pitta, 1995), which must still be executed either formally or informally by firms, large or small. This leads to looking at product management as a set of boundary spanning organizational capabilities, which must be performed by the firm in order to succeed in the marketplace. Product management can thus be thought of as a horizontal plane cutting through the traditional vertical specialties of the organization, bringing comprehensive management to product specific issues (Luck, 1969). This organizational level approach differs from the agency level approach of most of the product management literature.

Despite the importance of product management within the firm, there is no published scale to measure the strength of this relationship. Thus, for the purpose of this research, a 32-item survey instrument is proposed, based on the extant product management literature.

3. Measurement instrument development

In support of this study, the extant product management literature was synthesized, resulting in a measurement instrument consisting of 32 questions covering the breadth of the product management capabilities of the firm. The following outlines each set of questions and their relationship to product management capability.

At the core of product management is the product (or product line) strategy (Sands, 1979; Dawes and Patterson, 1988). The degree to which firms take a strategic planning orientation (Cossé and Swan, 1983; Lysonski *et al.*, 1995) should have an impact on firm performance. Product strategy determines which product categories the firm should be competing within and which categories it should relinquish. From an organizational perspective, product strategy has several boundary spanning implications. These include the cross functional responsibility for deciding which products to add and which to phase out (Eckles and Novotny, 1984) and the time horizon over which this takes place (for instance, short-term versus long-term) (Berek, 1998; Murphy and Gorchels, 1996; Katsanis, 2006; Wood and Tandon, 1994; Katsanis *et al.*, 1996). Companies with the capability of aligning their product strategy with the best alternative for managers (Bristow and Frankwick, 1994), should exhibit higher firm performance. Thus, as a measure of how the firm manages its *Product Strategy*, the following two questions (*PM1*, *PM28*) were used:

PM1. We have strong systems for deciding which products and/or service opportunities to pursue or which to phase-out.

PM28. We prefer to take a long-term approach to our product and/or service offerings rather than focusing merely on short-term profits.

One of the most challenging product management areas for most firms is pricing. Pricing touches all aspects of the firm's business model and is one of its most significant boundary spanning activities within organizations (McDaniel and Gray, 1980). Pricing decisions can be affected by the firm's strategic pricing philosophy (for instance cost-based versus value-based pricing) or whether the firm is involved in consumer or industrial sectors (Dawes and Patterson, 1988; Eckles and Novotny, 1984; Cummings *et al.*, 1984; Hise and Kelly, 1978). The ability to effectively set pricing involves strong internal and external communication for such issues as setting the base price, trade discounts, quantity discounts, price changes and pricing across channels (Gorchels, 2003; Eckles and Novotny, 1984; Cummings *et al.*, 1984; Lehmann and Winer, 2005). Good pricing policy should also be driven by good pricing analysis. Thus, as a measure of how the firm manages its Product Pricing, the following two questions (*PM2*, *PM3*) were used:

PM2. Our product and/or service pricing (for instance, base price, discount schedules, etc.) are clearly communicated to all stakeholders.

PM3. Our product and/or service pricing (for instance, base price, discount schedules, etc.) is determined via careful analysis and feedback from the marketplace.

Another significant product management activity is support for sales management. Classic product managers' work with sales management to set strategies and achieve plans (Katsanis and Pitta, 1999), while in SMEs these sales related activities tend to have a greater influence in marketing activities (Walsh, 2009). These can include contact with sales representatives, distributors, suppliers, and buyers (Sands, 1979; Coviello and Brodie, 2001) and should lead to the establishment of sales objectives (Eckles and Novotny, 1984), sales force quotas, sales force incentives, and allocation of sales resources. When looking specifically at the boundary spanning tasks of the firm, the product management activities tend to be generally related to sales support and motivation (Katsanis and Pitta, 1999; Gorchels, 2005) and can include such activities as product training, creating presentations/demos, technical/field support and special sales force requests (Gorchels, 2003; Murphy and Gorchels, 1996; Katsanis *et al.*, 1996; Cummings *et al.*, 1984; Cummings *et al.*, 1984; Petrini and Grub, 1973). Good sales management also has an after sale emphasis (Gray *et al.* 1998), which includes scrutinizing the effectiveness of marketing programs and/or performing win/loss analysis (Cossé and Swan, 1983). Thus, as a measure of how the firm manages its Product Sales Support, the following three questions (*PM4*, *PM5*, *PM6*) were used:

PM4. Our sales force or sales partners are consistently kept informed about our product and/or service strategy.

PM5. Our sales force or sales partners consistently receive up to date product and/or service support (for example, training, technical support, demonstrations and presentations).

PM6. Sales data is consistently analyzed and measured against marketing programs and forecasts.

Another well-documented role of product management within the organization is business analysis. This involves preparing the business case for new products or service, assessing the financial risk for new products, and evaluating the desirability of new products in the marketplace (Murphy and Gorchels, 1996; Cummings *et al.*, 1984). Thus, how the firm manages its product related Business Assessment is measured using the following question (*PM9*):

PM9. Our product and/or service plans are supported by strong business case analysis.

Closely related to the business case is product management's involvement in budgeting (Lysonski *et al.*, 1995; Cummings *et al.*, 1984; Hise and Kelly, 1978; Clewett and Stasch, 1975). This can range from assisting with the administration of marketing budgets, reviewing budget mix, launch budgets (Ledwith and O'Dwyer, 2008) and/or monitoring expenditures. In looking specifically at the product management activities associated with budgets, it is important to separate general marketing budgets from specific product or service related activities. For instance, is budgeting related generally to brand building or allocated at the product level? Also, tangentially related to budgeting is the product's overall profitability (Buell, 1975; Gorchels, 2003; Luck, 1969; Hise and Kelly, 1978). Thus, as a measure of how the firm manages it's Product Budgeting, the following two questions (*PM10*, *PM11*) are proposed:

PM10. Our marketing budget is allocated by product and/or service category rather than generally at the firm level.

PM11 We consistently monitor the profitability of each of our products and/or services.

Much of the product management literature highlights the monitoring of the product or service offering in an effort to understand the product-market dynamics. This can involve the monitoring of marketing programs to determine if they conform with and support the product plan. Other activities include measuring such activities as advertising effectiveness or monitoring commercial performance. The ultimate objective is to provide feedback to the product/service process within the firm for corrective action (Clewett and Stasch, 1975; Quelch *et al.*, 1992). Also, reviewing available product-related data such as product recalls, technical service data and customer satisfaction is critical to this process (Murphy and Gorchels, 1996; Gray *et al.*, 1998). Thus, how a firm conducts its *Internal* Monitoring activities is determined using the following three questions (*PM30*, *PM13*, *PM12*):

PM30. We consistently monitor the effectiveness of our promotional activities by product and/or service category.

PM13. Our firm consistently reviews customer comments, recalls and technical service data as input for future decisions about our products and/or services.

PM12 We consistently monitor our customer service by product and/or service category.

Monitoring the external environment including the regulatory or legal environment is another key product management activity (Katsanis and Pitta, 1999). It involves keeping abreast of potential conflicts between product strategies/programs and regional, national, or international laws/regulations. Issues such as the legitimacy of advertising claims, product labeling, or product recalls would be typical problems that require attention from the firm. The product management capability of the organization takes the role of resolving these issues, making recommendations and implementing action (Murphy and Gorchels, 1996; Katsanis, 2006; Cummings *et al.*, 1984; Quelch et al,. 1992). The following (*PM26*) is proposed to measure the External Monitoring activities of the firm's environment:

PM26. We consistently monitor our external environment (for example social, political, economic, regulatory) in order to avoid surprises with respect to our products or services.

One of the key external boundary spanning activities of the product management function is the integration of customer research into the product design and development process. This includes matching the needs of the market with the firm's technological opportunities and capabilities. This is the firm's boundary spanning capability of integrating customer learning into the product development process and typically involves primary research such as customer visits (Gorchels, 2003; Luck, 1969; Quelch *et al.*, 1992; Calantone *et al.*, 2002). As such, the following statement (*PM14*) is proposed to measure the Customer Integration aspect of product management:

PM14. Our firm consistently seeks customer input into the design of our products and/or services.

From a slightly different perspective, market research and competitive intelligence are mostly driven by secondary research. This external boundary spanning function is how firms search the industry for new ideas, trends, or phenomena (Katsanis, 1999). It involves keeping abreast of competitors, (Lysonski, 1985), and monitoring product diffusion (Calantone *et al.*, 2002) in an effort to anticipate changes in evolving market dynamics. It also involves the firm's capability to integrate competitive intelligence into new and existing products in the marketplace (Katsanis, 2006; Lysonski *et al.*, 1988; Quelch *et al.*, 1992). As such, the following three questions (*PM15*, *PM16*, *PM17*) are used to measure Competitive Intelligence:

PM15. Competitive research and analysis drives our product and/or service planning.

PM16. We consistently scan our industry for new ideas, trends or phenomena to improve our product and/or service offerings.

PM17. We systematically benchmark our product and/or service offerings against industry leaders.

Goal and objective setting is paramount to the success of the product management capability of the firm. This involves tradeoffs between internal resource allocation and the functional priorities of the organization. For instance, setting priorities to establish product-marketing objectives must be balanced with other functions such as cost accounting or research and development (Katsanis, 2006; Clewett and Stasch, 1975; Quelch *et al.*, 1992). Thus, having the systems to establish product, project, and personnel objectives should positively affect firm performance. As such, the following question (*PM18*) is proposed to measure Priority Setting:

PM18. We have well-established systems for setting priorities for our product or service offerings.

Forecasting has traditionally had a strong association with product management (Sands, 1979; Dominguez, 1971; Dawes and Patterson, 1988; McDaniel and Gray, 1980; Hise and Kelly, 1978). Forecasting involves a significant amount of boundary spanning activity inside and outside of the firm. It encompasses developing volume and revenue forecasts based on overall industry sales and market share by segment or product line (Cossé and Swan, 1983; Murphy and Gorchels, 1996; Katsanis, 2006; Quelch *et al.*, 1992). The Forecasting function is thus measured as follows (*PM19*, *PM20*):

PM19. We have well-established systems for forecasting and monitoring product and/or service unit sales.

PM20. We have well-established systems to monitor our market share by product and/or service category.

Communication and Coordination has a long history and is the subject of a significant amount of the product management literature (Sands, 1979; Dominguez, 1971; Berek, 1998; Dawes and Patterson, 1988; McDaniel and Gray, 1980). Communication has a broad scope and is difficult to fully specify. However, it can be broadly broken down into internal and external communications (Lysonski, 1985; Wood and Tandon, 1994). Examples of internal product management communication involve such things as conducting team meetings, fostering communication amongst diverse team members, and initiating and maintaining appropriate linkages between the product and project teams and/or functional departments. Examples of external product management communication can include contact with sales reps, distributors, suppliers, and buyers. These boundary-spanning processes require obtaining information and disseminating information, with the firm's product management capability providing a "filtering and facilitation" role (Wood and Tandon, 1994; p. 24). Fundamentally, communication involves transmitting information about the product/services from the external environment into the firm and *vice versa* (Katsanis, 2006; Lysonski *et al.*, 1988; Bristow and Frankwick, 1994; Clewett and Stasch, 1975). The Communication and Coordination function is measured as follows (*PM7*, *PM8*, *PM32*):

PM7. We foster communication about our products and/or services *within our firm* in order to generate product improvement ideas.

PM8. We foster communication about our products and/or services outside of our firm in order to generate product improvement ideas.

PM32. External information is consistently communicated to internal staff for decision making about our products and/or services.

Product development and R&D have a well-documented link with product management (Dawes and Patterson, 1988). The issues revolve around active participation within the firm to ensure that research and development plans are aligned with product strategy. The firm's product management capability needs to foster relationships within the organization throughout all of the stages of the development process (Luck, 1969). Tasks such as writing product requirements, determining product/service modifications, participating in the product conceptualization process, or assisting with product specifications are common themes (Murphy and Gorchels, 1996; Eckles and Novotny, 1984; Cummings *et al.*, 1984). Early involvement and integration of both marketing and R&D in the product development process should result in improved firm performance (Gorchels, 2003; Bristow and Frankwick, 1994; Calantone *et al.*, 2002). As such, the following three statements (*PM21*, *PM22*, *PM23*) are used to measure the boundary spanning roles of Marketing & Technical Integration:

PM21. We have well-established systems for involving business, marketing and technical personnel in our product and/or service efforts.

PM22. Marketing and technical personnel communicate effectively and work harmoniously when it comes to product and/or service issues.

PM23. Marketing and technical personnel participate equally in developing new product and/or service concepts.

Test marketing and pretest marketing (for example, simulated test market) are effective ways for firms to weed out questionable products prior to their market introduction/launch (Ledwith and O'Dwyer, 2008; Calantone *et al.*, 2002). This product management function is boundary spanning in nature, since it involves aspects of concept testing, product design, use/usability testing, as well as the more traditional aspects of test marketing (for example, brand name and price testing) (Murphy and Gorchels, 1996; Cummings *et al.*, 1984; Quelch *et al.*, 1992). More formal research studies also tend to prove or disprove internal notions about the product or service (Murphy and Gorchels, 1996). As such, the following three statements (*PM24*, *PM31*, *PM27*) are used to measure the Market Testing component of the product management function:

PM24. We consistently use pre-test marketing (for example simulated market tests) to establish the viability of new products and/or services.

PM31. Our products and services are mainly driven by market requests.

PM27. We frequently conduct formal research studies to prove or disprove our pre-conceived notions about our products and/or services.

Influence over the promotion and advertising decision-making process (Sands, 1979; Buell, 1975; Berek, 1998) is another classic function of product management. This can include many aspects of the promotional mix, including trade (and consumer) promotion, trade shows/exhibits, advertising copy, media selection, creating material for external audiences (for instance, blog/newsletter), and advertising. Consumer and industrial firms differ in how this responsibility is divided, with the former more highly involved in promotion and advertising than the latter (Gemmill, 1972; Murphy and Gorchels, 1996; Eckles and Novotny, 1984; Cummings *et al.*, 1984; Quelch *et al.*, 1992). The following question is thus used to measure the Cross-Functional Promotion and advertising approach of product management:

PM25. We take a team approach to the design and development of our promotional materials.

Last, value creation is another output of the product management capability of the firm. In the industrial product management model, technical depth and superiority tend to be more prevalent than in consumer product management systems (Murphy and Gorchels, 1996; Eckles and Novotny, 1984; Cummings *et al.*, 1984; Calantone *et al.*, 2002). As such, the following question is designed to measure whether the firm's product management functions are Technically Driven:

PM29. Our products and/or services are mainly driven by technical superiority.

It should be noted that no reverse questions were used in the development of the product management questionnaire. Given the exploratory nature of this research and the breadth of coverage of the product management capabilities of the firm, this was considered a reasonable assumption. Future work may use both positive and negative statements as a means of limiting halo bias.

Last, for the purposes of this research, performance will be measured using subjective, global performance measures as outlined in Appendix 1. These performance measures have been used extensively in SME research, driven largely by the reluctance of owner/managers to provide objective financial performance information (Wolff and Pett, 2006).

3.1 SME orientation and product management

Small- and medium-sized enterprises (SMEs) provide a unique challenge in the arena of product management. Despite the importance placed upon the marketing concept within SMEs, our understanding of the precise marketing activities and competencies that relate to firm performance remain under-developed (Walsh, 2009). Although the Procter and Gamble model of product management has made its way into some SMEs (Katsanis *et al.*, 1996), this formal model may not be very well suited for smaller organizations (Dawes and Patterson, 1988). Small- and medium-sized firms are often driven by daily emergencies and concerns, while frequently lacking adequate competitive analysis and new product development infrastructure (Woodcock *et al.*, 2000).

One of the key criticisms of many small- and medium-sized businesses is that they fail to translate solid research and development findings into successful innovations (O'Regan *et al.*, 2006). Research has shown that many small- and medium-sized businesses have strong technology related product management practices, such as product development, testing and analysis. However, the majority of SMEs lack the ability to properly perform many of the marketing related product management functions (for example, market research and market testing). These factors help differentiate successful and unsuccessful products (Huang *et al.*, 2002) and, along with customer intelligence, are positively related to improved innovation and profitability within small- and medium-sized enterprises (Verhees and Meulenberg, 2004). To overcome these innovation deficiencies, successful small- and medium-sized firms must devote a sufficient amount of energy to developing new products for current customers, broadening their markets/customer base and properly managing their current product portfolios (Smallbone *et al.*, 1995).

Research on innovation has also shown that improvements in SME's innovation practices can be achieved by beginning to structure organizations to allow for improved cross-functional and cross-disciplinary communication (Vermeulen, 2005). SMEs that are able to implement and nurture effective new product development practices appear to have a richer resource base and better formal or informal product management strategies (Karlsson and Olsson, 1998). Huang *et al.* (2002) research also provides some insight, since they conclude that certain product management practices such as market research, analysis, product creation and commercialization are the most crucial activities for product success. However, these activities are also believed to be the group of business activities that required the most improvement and attention (Karlsson and Olsson, 1998).

This evidence lends credibility and supports the importance of proper and effective product management capabilities within small- and medium-sized enterprises.

4. Methodology

The focus of this research is on small- and medium-sized enterprises (SMEs) engaged in manufacturing and professional/technical services. The geographic criteria used for this study was the Atlantic Provinces of Canada (Nova Scotia, New Brunswick, Prince Edward Island, and Newfoundland & Labrador).

Small- to medium-sized enterprises were defined as:

- having less than 250 employees
- less than CDN$ 50M in revenue; and
- are stand alone enterprises (that is, not subsidiaries of larger entities).

Sole proprietorships and micro-businesses were deselected by restricting this study to those businesses with greater than five employees. The study used key informants who were senior managers of SMEs (for example, General Managers through to CEOs).

The SME population was identified through the use of two prominent databases:

(1) Industry Canada's Canadian Company Capabilities (CCC) database; and

(2) the Canadian Business Directory (CBD).

The former is a self-reported, on-line database where Canadian companies freely share their corporate information, including senior company contact information, firm size, limited performance data, and key market information. The latter (CBD) is a commercially available database.

Records were also restricted to those businesses whose named contacts or officers were listed and telephone numbers were included. In total, 629 records in the CCC database and 628 in the CBD database were used. A number of methods were employed to detect common business records between the two datasets. Near matches were examined on an individual basis, such as subtle differences in spelling between the two systems. In total, 138 business records were positively matched between the two data sources, giving us a complete working data set of $N = 1{,}119$ distinct businesses.

The method of data collection was a self-reported on-line survey answered by senior company key informants between June and September 2008. Participants were asked to provide their opinion on a number of generic statements related to product management and firm performance. Firm performance was measured using a 9-item instrument (see Appendix 1). Questions were randomized using a seven-point Likert scale from "disagree completely" to "agree completely" based on how their firm has performed in the past, not how it hoped to perform in the future. Last, participants were asked briefly about their experience in their industry and their company, and what best described their management position within their organization.

This method resulted in 63 good responses ($N = 63$). This translated into a response rate of 5.63 percent of the total population ($N = 1{,}119$). Of these responses 50 (79.4 percent) were primarily producers of goods, while 13 (20.6 percent) were primarily suppliers of services. Of the respondents 44 (69.8 percent) were Chief Officers, Presidents or Vice Presidents of their companies. When senior managers were included this increased to 59 (93.7 percent) of respondents. Of these respondents 35 (55.61 percent) had greater than 15 years experience in their industry, while 28 (44.4 percent) had greater than 20 years experience.

5. Analysis and results

Given the small sample size collected ($N = 1{,}119$, $N = 63$) and for the purpose of exploratory research, linear regression was used to establish the relationships between product management (PM) measures and firm performance (PERF).

Results are as follows:

First, looking at the dependant variables for firm performance (nine items), we see that the Cronbach's Alpha of 0.84 indicates a good correlation between the measures of firm performance, well above the 0.70 "rule of thumb" standard (Hair *et al.*, 2006; Nunnally and Bernstein, 1994). This establishes that the performance measure is reliable and can be used as a dependable aggregate measure of firm performance. Thus, this nine-item performance measurement is used as the aggregate dependant variable for linear regression analysis.

The 32-item product management measures produce a Cronbach's Alpha of 0.949. This indicates that the proposed product management measures are internally consistent even given the large amount of variables considered. It is likely that this broad-based measurement encompasses several sub-constructs. However, without a larger sample size, principal component analysis (PCA) could not be performed. This suggests that the PM construct measurement merits further testing with a larger sample.

Next, significant sub-constructs of product management are reviewed. For the purposes of this analysis, only measures considered significant at a $p < 0.10$ level and with a Cronbach's Alpha greater than 0.60 (Nunnally and Bernstein, 1994) will be discussed (see Table I). Relaxing the constraint to $p < 0.10$ is considered a reasonable assumption. Although this may increase the chance of a type I error, the objective is to avoid a type II error for exploratory research (McBurney and White, 2003; Simon, 2005).

The two-item Product Pricing construct (*PM1*, *PM28*) shows both an acceptable Cronbach's Alpha (0.690), with an R^2 of 0.046 and significance of $p < 0.10$ when regressed against the aggregate performance measure. This supports the product management literature, which suggests that firms with a product management focus consider pricing to be a key product management capability.

Sales Support (*PM4*, *PM5*, *PM6*) also shows a good Cronbach's Alpha (0.701) and significance of $p < 0.10$, when regressed against performance. It also shows a strong R^2 of 0.139, indicating that this three-item measure has a strong relationship with firm performance in this sample. Of particular interest is one individual Sales Support measure (*PM5*), which shows a very strong R^2 of 0.185 with a significant correlation ($p < 0.001$). However, given the small sample size and the pilot nature of this investigation, the aggregate measure of Product Sales Support is considered a more reliable measure. This indicates that support to the sales function appears to be strongly related to SME firm performance.

Internal Product Monitoring (*PM12*, *PM13*, *PM30*) shows an acceptable Cronbach's Alpha (0.649), but this set of measures does not show a correlation with performance. Interestingly, External Product Monitoring (*PM26*), shows a direct relationship with firm performance, with an R^2 of 0.034 and a very strong correlation with performance ($p < 0.001$). If Product Monitoring is measured using both internal and external

Construct description	Items	No. of items	Cronbach's Alpha	R^2	Sig.
Firm performance	PERFORM1-6, A-B Export	9	0.840	n/a	n/a
Product management orientation (all)	*PM1-32*	32	0.949	n.s.	n.s.
Product strategy	*PM1*, *PM28*	2	n.s	n.s.	n.s.
Product pricing	*PM2*, *PM3*	2	0.690	0.046	*
Product sales support	*PM4*, *PM5*, *PM6*	3	0.701	0.139	**
Sales support	*PM5*	1	n/a	0.185	***
Product budgeting	*PM10*, *PM11*	2	n.s	n.s.	n.s.
Internal product monitoring	*PM12*, *PM13*, *PM30*	3	0.649	n.s.	n.s.
External product monitoring	*PM26*	1	n/a	0.034	***
Monitoring	*PM12*, *PM13*, *PM26*, *PM30*	3	0.734	n.s.	n.s.
Competitive intelligence	*PM15*, *PM16*, *PM17*	3	0.715	n.s.	n.s.
Forecasting	*PM19*, *PM20*	2	0.684	0.045	*
Communication and coordination	*PM7*, *PM8*, *PM32*	3	0.709	n.s.	n.s.
Marketing & technical integration	*PM21*, *PM22*, *PM23*	3	0.616	n.s.	n.s.
Market testing	*PM24*, *PM27*, *PM31*	3	n.s	n.s.	n.s.
Technical superiority	*PM29*	1	n/a	0.113	***
Non-significant PMO[a]	*PM_NON_SIG*	23	0.927	n.s.	n.s.

Notes: [a]Remaining PM measures with a non-significant relationships with firm performance (using the criteria $p < 0.10$); $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$

Table I.
Reliability statistics

measures, the Cronbach's Alpha improves to 0.734, indicating an improvement in the consistency of the measurement, though it fails to improve in significance when regressed against performance in this sample. Thus, Product Monitoring as a construct seems to gain consistency by including both internal and external dimensions indicating that it merits further investigation in future research. This may shed some light on some of the unexplained interaction between internal and external marketing activities of the firm (Walsh, 2009).

Forecasting (*PM19*, *PM20*) is well established in the product management literature as a specific capability. In our sample, it displays a reasonable Cronbach's Alpha (0.684) with an R^2 of 0.045 and significance of $p < 0.10$, when regressed against performance. This two-item measure indicates that strong forecasting capability may have a role in the prediction of firm performance.

One final single measure of Product Management, namely Technical Superiority displays significance in this sample with an R^2 of 0.113, which indicates a strong relationship with performance ($p < 0.001$). This suggests that the high performing firms in this sample seemed to differentiate themselves via technical superiority. This measure could be an artifact of this particular sample, since it is weighted towards industrial firms within a specific geographic area. This should be investigated in more detail in the future research.

Several other constructs showed internal consistency with Cronbach's Alpha > 0.60, including Competitive Intelligence, Communication and Coordination and Marketing & Technical Integration. This indicates that even with this small under-determined dataset, these constructs showed consistency and merit further investigation in a larger study.

6. Discussion

This preliminary study and analysis has highlighted several issues for consideration. These can be summarized into three general areas, namely:

(1) specific product management measures or constructs;

(2) selection of the target population; and

(3) relationship to other firm orientations.

First, specific measurement questions and sub-constructs were found to be important and significant at the exploratory level. These include Product Pricing, Sales Support, Forecasting and Technical Superiority. These constructs are all well established in the product management literature and this study indicates that these measures show both internal consistency and correlation with firm performance. This is academically important since it indicates that a relationship exists between the product management literature and firm performance. However, this relationship is provisional, given the nature of this exploratory research, which included relaxing traditional statistical constraints to $p < 0.10$, considered reasonable for investigative research (McBurney and White, 2003). Yet, this does encourage the use of these measures in a larger, more broadly based study.

Also of note, are several other sub-constructs which showed internal consistency, but not a correlation with firm performance. These included Monitoring (both internal and external), Competitive Intelligence, Communication & Coordination and Marketing & Technical Integration. These constructs are also well established in the product management literature and their internal consistency suggests that these constructs deserve further investigation in future studies.

A few individual product management measures showed strong significance ($p < 0.001$) on their own and included Sales Support (*PM5*) and Technical Superiority (*PM29*). Although *PM5* showed both strength and significance, it appears to be related to a higher-level construct of Product Sales Support (*PM4-5-6*), which is supported by the extant literature. On the other hand, *PM29* shows both strength and significance, which could be a reflection of the chosen population (or sample) in this pilot dataset. A larger study where controls for industry can be employed may shed more light on this relationship.

Second, selection of the target population is important for future studies. This pilot study was conducted using a convenience sample (SMEs in Atlantic Canada) in conjunction with a policy research study for a Canadian Federal Agency (the Atlantic Canada Opportunities Agency – ACOA). This introduced geographic and selection bias (for instance, combining manufacturing & service firms), along with a small sample size ($N = 63$ out of $N = 1,119$). For future studies, a larger and more homogeneous population is recommended. For instance, using a broader population of Canadian SMEs, where larger population sizes are available, should reduce or eliminate these biases. This should also allow for the use of controls, such as industry sectors.

Last, future studies could benefit from measuring product management in conjunction with other well established firm orientations. For instance, the market orientation (MO) construct (Kohli *et al.*, 1993; Jaworski and Kohli, 1993; Deshpandé *et al.*, 1993; Narver and Slater, 1990) has a long and rich body of academic research, which may add credibility and understanding to the proposed measures of product management. In a similar vein, firm-level innovativeness (Calantone *et al.*, 2002; Gatignon and Xuereb, 1997; Paladino, 2007) could add to the understanding of product management, since it spans the firm's traditional boundaries between marketing and innovation orientations.

7. Limitations and future research

This study is exploratory in nature and its primary aim is to assist with hypothesis generation rather than providing hypothesis testing. The author is careful in drawing conclusions based on this limited study; thus, the analysis provided is considered preliminary in nature. It offers good initial evidence of construct validity of the proposed product management concept, while providing insight into the boundary spanning capabilities of the firm and their relationship to performance. It does this by examining a small, under-determined set of Atlantic SMEs in both the manufacturing and professional/technical services sectors.

There are several limitations to this study, which will need to be resolved in future research. These include the geographic nature of the population, the small sample size, the treatment of two distinct industry types as homogeneous, the broad nature of the product management construct and the lack of relationship to other well researched firm orientations. All of these issues should be addressed in future research. Although this study does not distinguish between firms involved in manufacturing and professional/technical services, the literature does make this distinction (Eckles and

Novotny, 1984; Cummings *et al.*, 1984; Walsh, 2009; Coviello and Brodie, 2001). This might well have an effect on how firms configure their resources and capabilities.

The broad nature of the product management construct can also be resolved through its relationship with other firm orientations such as market, entrepreneurial and/or innovation orientations. From the literature review it is clear that product management is a broad construct, which encompasses several factors, one or more of which are likely measuring alternate constructs of firm orientation. This issue should be clarified in future research using a larger population and more sophisticated methods of analysis, including factor analysis and/or structural equation modeling. This could allow for the examination of correlations between individual measurements and possible latent factors.

Regardless, it is clear from this preliminary analysis that the proposed measures of product management provide some insight into firm performance. Further research will be needed using a larger sample of SMEs to establish which measures of product management provide predictability of firm performance.

8. Conclusions

Product management is a wide-ranging construct whose activities span the internal functional areas of the firm and its external environment. According to Olson *et al.* (2005), researchers should concentrate more on understanding the antecedents of marketing practices in order to predict performance and inter-functional coordination. How work is coordinated within the firm, should receive more attention in the area of strategy implementation research. Although organizations large and small perform many of these boundary-spanning activities, those that excel at their integration should perform better and more easily adapt to their competitive environments. While firms still rely on either formal or informal product management practices, researchers have not tested this in any empirical way.

This pilot study has many purposes. First, it offers preliminary validation of product management measures related to firm performance with a small sample of regional SMEs. This is academically important since the literature has thus far not linked the construct of product management to firm performance. Second, the study's focus on SMEs adds to the product management literature, which is either focused on large enterprises or does not distinguish between small, medium, or large organizations. Third, it establishes the reliability of self-reported objective measures of performance with a small sample of SMEs. This is important as it establishes these objective performance measures as valid for future research in this area. Fourth, the study seeks to establish the degree to which boundary spanning product management practices are currently employed by SMEs. Last, this research may have practical significance and managerial implications if product management practices can be related to firm performance. This could potentially lead to an increased understanding of how to allocate resources in order to improve firm performance. It may also have policy implications related to programs that support commercialization and innovation within SMEs.

In conclusion, this small sample of 63 Atlantic Canadian SMEs provides support for much of the literature on product management. The researcher developed product management measures did highlight some correlation with constructs from the literature and firm performance. These included Product Pricing, Sales Support, Forecasting and Technical Superiority. It also highlighted several constructs that merit further investigation, including Monitoring, Competitive Intelligence, Communication & Coordination and Marketing & Technical Integration. This is encouraging and highlights the need for a larger, more broadly based study.

Most significantly, this research introduces the concept of boundary spanning, product management capabilities and their relationship to firm performance.

References

Berek, R.L. (1998), "Product management a work in progress", *Pharmaceutical Executive*, Vol. 18 No. 3, p. 60.

Bristow, D. and Frankwick, G. (1994), "Product managers' influence tactics in marketing strategy development and implementation", *Journal of Strategic Marketing*, Vol. 2 No. 3, p. 211.

Buell, V. (1975), "The changing role of the product manager in consumer goods companies", *The Journal of Marketing*, Vol. 39 No. 3, p. 3.

Calantone, R.J., Cavusgil, S. and Zhaob, Y. (2002), "Learning orientation, firm innovation capability, and firm performance", *Industrial Marketing Management*, Vol. 31 No. 6, p. 515.

Chaston, I. and Mangles, T. (1997), "Core capabilities as predictors of growth potential in small manufacturing firms", *Journal of Small Business Management*, Vol. 35 No. 1, p. 47.

Clewett, R. and Stasch, S. (1975), "Shifting role of the product manager", *Harvard Business Review*, Vol. 53 No. 1, p. 65.

Connolly, K.E. (2002), "Growth by design: product management keeps financial giant nimble", *Design Management Journal*, Vol. 13 No. 2, p. 24.

Cossé, T. and Swan, J. (1983), "Strategic marketing planning by product managers – room for improvement?", *The Journal of Marketing*, Vol. 47 No. 3, p. 92.

Coviello, N. and Brodie, R. (2001), "Contemporary marketing practices of consumer and business-to-business firms: how different are they", *The Journal of Business Industrial Marketing*, Vol. 16 No. 5, p. 382.

Cummings, T.W., Jackson, D.W. and Ostrom, L.L. (1989), "Examining product managers' job satisfaction and performance using selected organizational behavior variables", *Journal of the Academy of Marketing Science*, Vol. 17 No. 2, p. 147.

Cummings, T.W., Jackson, D.W. and Ostrom, L.L. (1984), "Differences between industrial and consumer product managers", *Industrial Marketing Management*, Vol. 13 No. 3, p. 171.

Dawes, P.L. and Patterson, P.G. (1988), "The performance of industrial and consumer product managers", *Industrial Marketing Management*, Vol. 17 No. 1, p. 73.

Deshpandé, R., Farley, J. and Webster, F. (1993), "Corporate culture, customer orientation, and innovativeness in Japanese firms: a quadrad analysis", *Journal of Marketing*, Vol. 57 No. 1, p. 23.

Dominguez, G. (1971), *Product Management*, American Management Association, New York, NY.

Drucker, P.F. (1954), *The Practice of Management*, Harper & Row, New York, NY.

Eckles, R. and Novotny, T.J. (1984), "Industrial product managers: authority and responsibility", *Industrial Marketing Management*, Vol. 12 No. 2, p. 71.

Gatignon, H. and Xuereb, J. (1997), "Strategic orientation of the firm and new product performance", *Journal of Marketing Research*, Vol. 34 No. 1, p. 77.

Gemmill, R.G. (1972), "The product manager as an influence agent", *Journal of Marketing*, Vol. 36 No. 1, p. 26.

Giese, T.D. and Wiesenberger, T.M. (1982), "Product manager in perspective", *Journal of Business Research*, Vol. 10 No. 3, p. 267.

Gorchels, L. (2003), "Transitioning from engineering to product management", *Engineering Management Journal*, Vol. 15 No. 4, p. 40.

Gorchels, L. (2005), *The Product Manager's Handbook*, McGraw-Hill, New York, NY.

Gray, B., Matear, S., Boshoff, C. and Matheson, M. (1998), "Developing a better measure of market orientation", *European Journal of Marketing*, Vol. 32 No. 9, p. 884.

Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E. and Tatham, R.L. (2006), *Mulitvariate Data Analysis*, Pearson Prentice Hall, Upper Saddle River, NJ.

Hise, R.T. and Kelly, P.J. (1978), "Product management on trial", *Journal of Marketing*, Vol. 42 No. 4, p. 28.

Huang, X., Soutar, G.N. and Brown, A. (2002), "New product development processes in small and medium-sized enterprises: some Australian evidence", *Journal of Small Business Management*, Vol. 40 No. 1, p. 27.

Jaworski, B. and Kohli, A. (1993), "Market orientation: antecedents and consequences", *The Journal of Marketing*, Vol. 57 No. 3, p. 53.

Karlsson, C. and Olsson, O. (1998), "Product innovation in small and large enterprises", *Small Business Economics*, Vol. 10 No. 1, p. 31.

Katsanis, L. (1999), "Some effects of changes in brand management systems: issues and implications", *International Marketing Review*, Vol. 16 No. 6, p. 518.

Katsanis, L. (2006), "An assessment of professional training for product managers in the pharmaceutical industry", *The Journal of Product & Brand Management*, Vol. 15 No. 5, p. 324.

Katsanis, L. and Pitta, D.A. (1995), "Punctuated equilibrium and the evolution of the product manager", *The Journal of Product & Brand Management*, Vol. 4 No. 3, p. 49.

Katsanis, L. and Pitta, D.A. (1999), "The performance appraisal process of pharmaceutical product managers in Canada: an empirical study", *Journal of Product & Brand Management*, Vol. 8 No. 6, p. 463.

Katsanis, L.P., Laurin, J. and Pitta, D.A. (1996), "How should product managers' job performance be evaluated in emerging product management systems?", *Journal of Product & Brand Management*, Vol. 5 No. 6, p. 5.

Kohli, A., Jaworski, B. and Kumar, A. (1993), "MARKOR: a measure of market orientation", *Journal of Marketing Research*, Vol. 30 No. 4, p. 467.

Ledwith, A. and O'Dwyer, M. (2008), "Product launch, product advantage and market orientation in SMEs", *Journal of Small Business and Enterprise Development*, Vol. 15 No. 1, p. 96.

Lehmann, D. and Winer, R. (2005), *Product Management*, McGraw-Hill/Irwin, Boston, MA.

Luck, D.J. (1969), "Interfaces of a product manager", *Journal of Marketing*, Vol. 33 No. 4, p. 32.

Lysonski, S. (1985), "A boundary theory investigation of the product manager's role", *The Journal of Marketing*, Vol. 49 No. 1, p. 26.

Lysonski, S. and Woodside, A. (1989), "Boundary role spanning behavior, conflicts and performance of industrial product managers", *Journal of Product Innovation Management*, Vol. 6 No. 3, p. 169.

Lysonski, S., Levas, M. and Lavenka, N. (1995), "Environmental uncertainty and organizational structure: a product management perspective", *The Journal of Product & Brand Management*, Vol. 4 No. 3, p. 7.

Lysonski, S., Singer, A. and Wilemon, D. (1988), "Coping with environmental uncertainty and boundary spanning in the product manager's role", *The Journal of Services Marketing*, Vol. 2 No. 4, p. 15.

McBurney, D.H. and White, T.L. (2003), *Research Methods*, Wadsworth Publishing, Belmont, CA.

McDaniel, C. and Gray, D.A. (1980), "The product manager", *California Management Review*, Vol. 23 No. 1, p. 87.

Murphy, W. and Gorchels, L. (1996), "How to improve product management effectiveness", *Industrial Marketing Management*, Vol. 25 No. 1, p. 47.

Narver, J. and Slater, S. (1990), "The effect of a market orientation on business profitability", *The Journal of Marketing*, Vol. 54 No. 4, p. 20.

Nunnally, J. and Bernstein, I. (1994), *Psychometric Theory*, McGraw Hill, New York, NY.

Olson, E., Slater, S. and Hult, G. (2005), "The performance implications of fit among business strategy, marketing organization structure, and strategic behavior", *The Journal of Marketing*, Vol. 69 No. 3, p. 49.

O'Regan, N., Ghobadian, A. and Sims, M. (2006), "Fast tracking innovation in manufacturing SMEs", *Technovation*, Vol. 26 No. 2, p. 251.

Paladino, A. (2007), "Investigating the drivers of innovation and new product success: a comparison of strategic orientations", *The Journal of Product Innovation Management*, Vol. 24 No. 6, p. 534.

Pelham, A. (1997), "Mediating influences on the relationship between market orientation and profitability in small industrial firms", *Journal of Marketing Theory and Practice*, Vol. 5 No. 3, p. 55.

Petrini, B.F. and Grub, P.G. (1973), "Product management in high technology defence industrial marketing", *California Management Review*, Vol. 15, p. 138.

Quelch, J., Farris, A. and Oliver, J. (1992), "The product management audit: design and survey findings", *The Journal of Product & Brand Management*, Vol. 1 No. 4, p. 21.

Sands, S. (1979), "Is the product manager obsolete?", *Business Quarterly*, Vol. 44 No. 3, p. 30.

Simon, S. (2005), "Stats: what is an alpha level?", available at: www.childrens-mercy.org/stats/definitions/alpha.htm (accessed 8 August, 2010).

Smallbone, D., Leig, R. and North, D. (1995), "The characteristics and strategies of high growth SMEs", *International Journal of Entrepreneurial Behaviour Research*, Vol. 1 No. 3, p. 44.

Strieter, J., Gupta, A.K., Raj, S. and Wilemon, D. (1999), "Product management and the marketing of financial services", *International Journal of Bank Marketing*, Vol. 342, p. 354.

Tyagi, R.K. and Sawhney, S.S. (2010), "High-performance product management: the impact of structure, process, competencies, and role definition", *The Journal of Product Innovation Management*, Vol. 27 No. 1, p. 83.

Venkatesh, A. and Wilemon, D.L. (1976), "Interpersonal influence in product management", *Journal of Marketing*, Vol. 40 No. 4, p. 33.

Verhees, F. and Meulenberg, M. (2004), "Market orientation, innovativeness, product innovation, and performance in small firms", *Journal of Small Business Management*, Vol. 42 No. 2, p. 134.

Vermeulen, M. (2005), "Uncovering barriers to complex incremental product innovation in small and medium-sized financial services firms", *Journal of Small Business Management*, Vol. 43 No. 4, p. 432.

Walsh, M.F. (2009), "The role of the marketing function in small and medium sized enterprises", *Journal of Small Business and Enterprise Development*, Vol. 16 No. 4, p. 569.

Wolff, J.A. and Pett, T.L. (2006), "Small-firm performance: modeling the role of product and process improvements", *Journal of Small Business Management*, Vol. 44 No. 2, p. 268.

Wood, V. and Tandon, S. (1994), "Key components in product management success (and failure)", *The Journal of Product & Brand Management*, Vol. 3 No. 1, p. 19.

Woodcock, D., Mosey, S. and Wood, T. (2000), "New product development in British SMEs", *European Journal of Innovation Management*, Vol. 3 No. 4, p. 212.

Appendix 1 – Performance measurement

Table AI.

PERFORM1	Our sales growth was ___________ when compared to our competitors
PERFORM2	Our profit growth was ___________ when compared to our competitors
PERFORM3	Our employment growth was ___________ when compared to our competitors
PERFORM4	The overall performance of the business met expectations last year
PERFORM5	The overall performance of the business last year exceeded that of our major competitors
PERFORM6	Top management was very satisfied with the overall performance of the business last year
PERFORMA	Over the past 3 years, our sales growth on average:
PERFORMB	Generally, our firm's ability to raise capital is:
Export	Exports represent (percentage of revenue; 0, 25, 50, 75, 100)

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